CSCE 763: Digital Image Processing
Homework #3

Instructor: Spring 2026
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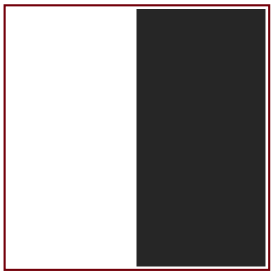
Due: 1:15 pm EST, Tuesday, Feb 17

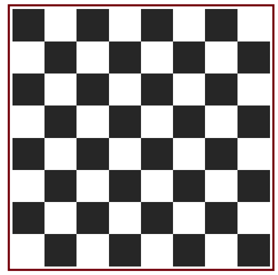
Table of Contents

Problem 1: Histogram and Blurring (20 pts)

Two $N \times N$ images shown below are quite different, but their histograms are the same. Suppose that each image is blurred with a 3×3 averaging mask. Please ignore the low quality of the images. You can assume all the black regions have a uniform intensity of 0, and the white regions have a uniform intensity of 255.



(a) Half-and-half



(b) Chessboard

(a) Would the histogram of the blurred images still be equal? Explain.

(b) If your answer is no, sketch the two histograms.

(c) **Bonus question:** Discuss how the histogram of the blurred image (b) is affected by different image size N. (5 pts)

Step 1: Analyze the Original Histograms

Both images have $N^2/2$ pixels at 0 and $N^2/2$ at 255, so their histograms are identical two-impulse distributions.
Assumption: The chessboard is a *1-pixel checkerboard* where pixel (i, j) has intensity 255 if $i+j$ is even, and 0 if $i+j$ is odd. If the squares were larger blocks, interior pixels would be unaffected by 3×3 averaging, and the histograms would remain similar.

Step 2: Analyze Blurring of Image (a) — Half-and-Half

Interior pixels on either side have all 9 neighbors the same color, so they remain at 0 or 255. Only pixels within 1 column of the vertical boundary are affected — the 3×3 window straddles both halves, producing intermediate values. This affects only $3N$ out of N^2 pixels, so the histogram retains two large spikes at 0 and 255 with a small spread of intermediate values.

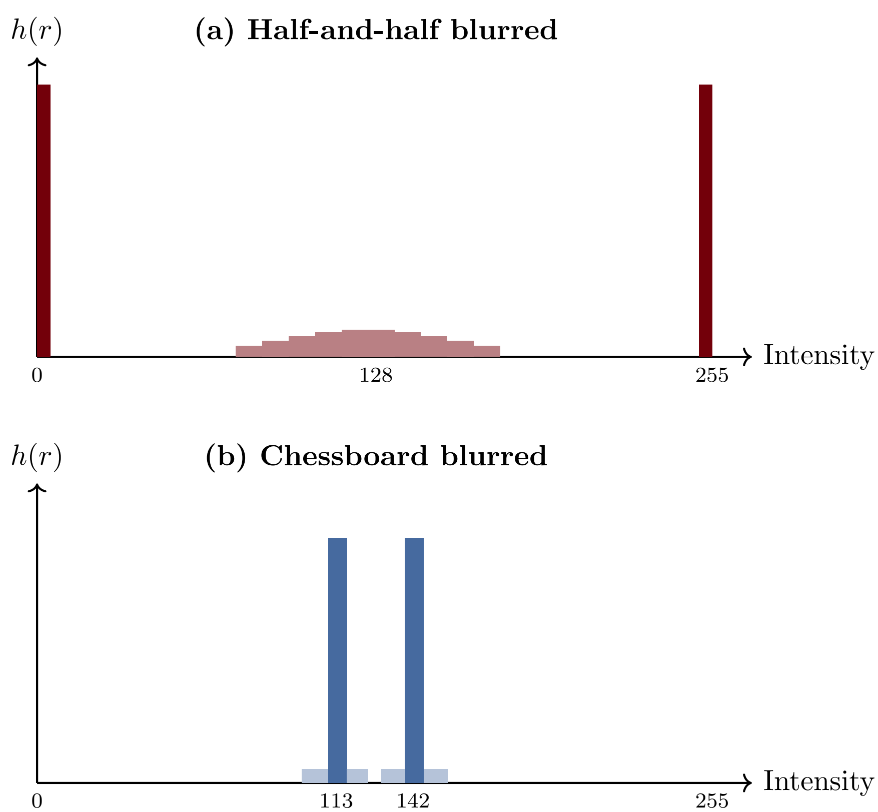
Step 3: Analyze Blurring of Image (b) — Chessboard

Every pixel is adjacent to opposite-intensity pixels. For any interior pixel, the 3×3 neighborhood has 5 same-color and 4 opposite-color pixels (4 corners match, 4 edges differ). If the center is white: average $= (5 \times 255)/9 = 1275/9 \approx 141.7$. If the center is black: average $= (4 \times 255)/9 = 1020/9 \approx 113.3$. **Every** pixel is affected, and the histogram collapses to two spikes near 142 and 113.

Part (a): Are the Histograms Still Equal?

No. Image (a) retains most pixels at 0 and 255 with a thin strip of intermediate values. Image (b) has *no* pixels at 0 or 255; the entire image shifts to two clusters near 113 and 142. The histograms differ because the 3×3 mask interacts with the *spatial structure*, not just the intensity distribution.

Part (b): Sketch of the Two Blurred Histograms



Part (c) Bonus: Effect of Image Size N on Chessboard Histogram

The $(N-2)^2$ interior pixels produce values ≈ 142 or ≈ 113, while border pixels (with zero-padded neighbors) produce different values shifted toward 0. The border fraction is $\sim 4/N$, so as N increases, the histogram converges to two sharp spikes at 142 and 113. For small N, border effects dominate and the histogram is more spread out.

Results

(a) No, the histograms of the blurred images are not equal. Image (a) retains large spikes at 0 and 255 with few intermediate values, while image (b) collapses entirely to two narrow peaks near 113 and 142.
(b) See histogram sketches above. Image (a) has a bimodal histogram concentrated at the extremes; image (b) has a bimodal histogram concentrated near the center.
(c) Bonus: As $N \to \infty$, the border-pixel fraction $\to 0$ and the chessboard histogram converges to two delta functions at ≈ 113 and ≈ 142. For small N, border effects widen and shift the peaks.

Problem 2: Spatial Filtering (20 pts)

Consider spatial filtering.

(a) Suppose that you filter an image $f(x, y)$, with a spatial filter mask $w(x, y)$, using convolution, as defined in Eq. (3.4-2), where the mask is smaller than the image in both spatial directions. Show the important property that, if the coefficients of the mask sum to zero, then the sum of all the elements in the resulting convolution array (filtered image) will be zero also (you may ignore computational inaccuracies). Also, you may assume that the border of the image has been padded with the appropriate number of zeros. (10 pts)

$$w(x, y) \otimes f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t)\, f(x - s, y - t) \tag{3.4-2}$$

(b) Would the result to (a) be the same if the filtering is implemented using correlation, as defined in Eq. (3.4-1)? (10 pts)

$$w(x, y) \odot f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t)\, f(x + s, y + t) \tag{3.4-1}$$

Part (a): Convolution with Zero-Sum Mask

Let $g(x, y) = w(x, y) \otimes f(x, y)$ denote the filtered image. We want to show that if $\displaystyle\sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) = 0$, then $\displaystyle\sum_{x} \sum_{y} g(x, y) = 0$.

Step 1: Write the Sum of All Output Pixels

$$\sum_{x} \sum_{y} g(x, y) = \sum_{x} \sum_{y} \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t)\, f(x - s, y - t)$$

Step 2: Exchange the Order of Summation

Since all sums are finite, we can exchange the order:

$$\sum_{x} \sum_{y} g(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) \left[\sum_{x} \sum_{y} f(x - s, y - t) \right]$$

Step 3: Evaluate the Inner Sum

Consider the inner sum $\sum_x \sum_y f(x-s, y-t)$. Substituting $u = x - s$, $v = y - t$ (a change of summation index), and noting that f is zero-padded (i.e., $f(u,v) = 0$ outside the image), the summation range covers all positions where f is nonzero regardless of the shift (s,t). Therefore:

$$\sum_x \sum_y f(x-s, y-t) = \sum_u \sum_v f(u,v) = S_f \quad \text{(constant, independent of } s,t\text{)}$$

where S_f is the sum of all pixel values in the original image.

Step 4: Factor and Conclude

Substituting back:

$$\sum_x \sum_y g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) \cdot S_f = S_f \cdot \underbrace{\sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)}_{=0 \text{ (given)}} = 0 \qquad \blacksquare$$

Part (b): Correlation with Zero-Sum Mask

For correlation, $g'(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x+s, y+t)$. The same argument applies:

$$\sum_x \sum_y g'(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) \left[\sum_x \sum_y f(x+s, y+t) \right]$$

With the substitution $u = x + s$, $v = y + t$, the inner sum is again S_f (independent of s,t):

$$\sum_x \sum_y g'(x,y) = S_f \cdot \sum_{s,t} w(s,t) = S_f \cdot 0 = 0 \qquad \blacksquare$$

Yes, the result is the same for correlation. The sign change in the index ($x - s$ vs. $x + s$) does not affect the argument because the inner sum over a shifted version of f always equals S_f under zero-padding.

Results

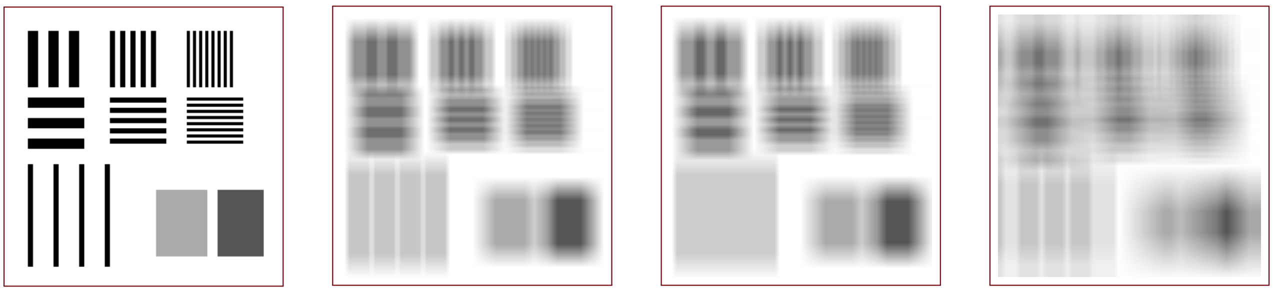
(a) If the mask coefficients sum to zero, i.e., $\sum_{s,t} w(s,t) = 0$, then the sum of all elements in the convolution output is:

$$\sum_{x,y} g(x,y) = S_f \cdot \sum_{s,t} w(s,t) = 0 \qquad \blacksquare$$

(b) Yes, the result is identical for correlation. The proof is the same because the inner summation $\sum_{x,y} f(x+s, y+t) = S_f$ is independent of the shift direction. $\blacksquare$

Problem 3: Averaging Mask and Bar Merging (20 pts)

The three images shown were blurred using square averaging masks of sizes $n = 23$, 25, and 45, respectively. The vertical bars on the left lower part of (a) and (c) are blurred, but a clear separation exists between them. However, the bars have merged in image (b), in spite of the fact that the mask that produced this image is significantly smaller than the mask that produced image (c).



Original **(a)** $n = 23$ **(b)** $n = 25$ **(c)** $n = 45$

Explain the reason for this. The vertical bars are 5 pixels wide, 100 pixels high, and their separation is 20 pixels. (Hints: you can simplify the problem by considering a single scan line through the bars in the image.)

Step 1: Simplify to a 1D Problem

Following the hint, consider a single horizontal scan line through the bars. The 1D intensity profile is periodic with bar width 5 pixels (intensity 255), gap width 20 pixels (intensity 0), and **period** $T = 5 + 20 = 25$ **pixels**. The $n \times n$ averaging mask reduces to a 1D moving average of width n.

Step 2: Effect of a Moving Average on a Periodic Signal

A 1D moving average of width n computes:

$$g(x) = \frac{1}{n} \sum_{k=0}^{n-1} f(x-k)$$

This sums n consecutive pixels. For a periodic signal with period T, the key observation is: if n is an exact multiple of T, then the window always contains the same number of "on" (bar) and "off" (gap) pixels regardless of position, making the output **constant**.

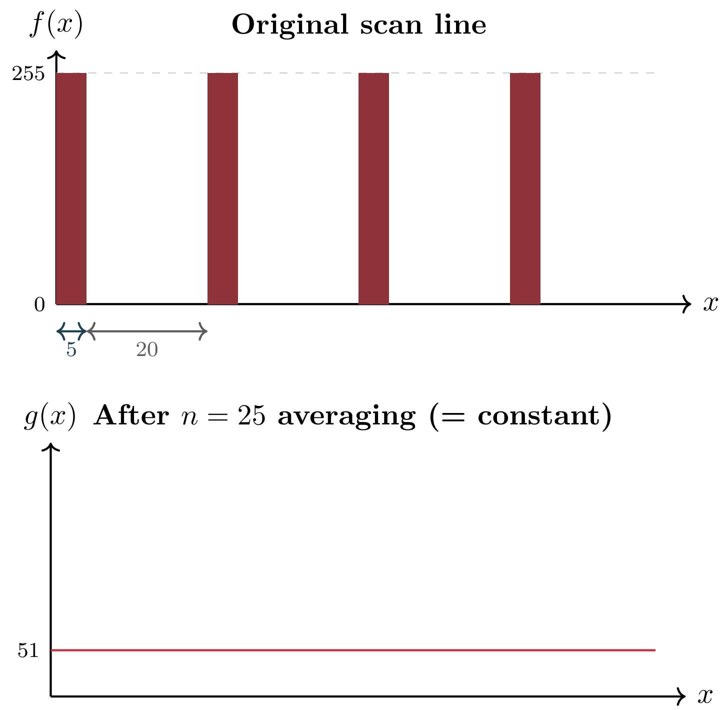
Step 3: Why $n = 25$ Causes Complete Merging

Since $T = 25$ and $n = 25 = 1 \times T$, the averaging window spans **exactly one full period**. At every position, the window contains exactly 5 bar pixels (at 255) and 20 gap pixels (at 0):

$$g(x) = \frac{5 \times 255 + 20 \times 0}{25} = \frac{1275}{25} = 51 \quad \text{(constant)}$$

The output is a uniform gray value of 51 along the entire scan line. All spatial variation is destroyed, and the bars **merge completely**.

Visualization: 1D Scan Line Analysis



Step 4: Why $n = 23$ and $n = 45$ Preserve Separation

$n = 23 < T$**:** The window is smaller than one period, so the number of bar pixels it covers varies with position, preserving the intensity modulation.
$n = 45 \neq kT$**:** Since $45 = 25+20$ is not a multiple of T, the window alternates between covering 1 and 2 bars. The output is more blurred but spatial variation persists.

Results

The bars merge completely when $n = 25$ because the averaging mask width **exactly matches the period** of the bar pattern ($T = 5 + 20 = 25$ pixels). When $n = T$, the moving average window always contains the same number of bar and gap pixels at every position, producing a constant output and destroying all spatial variation.
For $n = 23 < T$ and $n = 45 \neq kT$, the window size does not match the period, so the number of bar pixels in the window varies with position, preserving the intensity modulation and keeping the bars distinguishable. $\blacksquare$

Problem 4: Averaging and Laplacian Order (20 pts)

In a given application an averaging mask is applied to input images to reduce noise, and then a Laplacian mask is applied to enhance small details. Would the result be the same if the order of these operations were reversed?

Step 1: Express Both Operations as Convolutions

Let w_{avg} denote the averaging mask and w_{lap} denote the Laplacian mask. Both are linear spatial filters, so both operations are convolutions:

$$g_1 = w_{\text{lap}} \otimes (w_{\text{avg}} \otimes f) \qquad \text{(average first, then Laplacian)}$$

$$g_2 = w_{\text{avg}} \otimes (w_{\text{lap}} \otimes f) \qquad \text{(Laplacian first, then average)}$$

Step 2: Apply Associativity and Commutativity of Convolution

Convolution is both **associative** and **commutative**:

$$w_1 \otimes (w_2 \otimes f) = (w_1 \otimes w_2) \otimes f = (w_2 \otimes w_1) \otimes f = w_2 \otimes (w_1 \otimes f)$$

Applying this to our operations:

$$g_1 = w_{\text{lap}} \otimes (w_{\text{avg}} \otimes f) = (w_{\text{lap}} \otimes w_{\text{avg}}) \otimes f$$

$$g_2 = w_{\text{avg}} \otimes (w_{\text{lap}} \otimes f) = (w_{\text{avg}} \otimes w_{\text{lap}}) \otimes f$$

Since $w_{\text{lap}} \otimes w_{\text{avg}} = w_{\text{avg}} \otimes w_{\text{lap}}$ (commutativity), we have:

$$g_1 = g_2$$

Results

Yes, the result is the same regardless of the order. Both the averaging mask and the Laplacian mask are linear, shift-invariant operators (i.e., convolutions). Since convolution is associative and commutative:

$$w_{\text{lap}} \otimes (w_{\text{avg}} \otimes f) = w_{\text{avg}} \otimes (w_{\text{lap}} \otimes f)$$

The mathematical output is identical. ■

Problem 5: Convolution Filtering (20 pts)

Given the following 3×3 filter, perform the image filtering using **convolution**. Give the **FULL** filtering result for each filtering process. (Hint: you need to assume an appropriate zero mapping.)

$$w = \begin{bmatrix} -1 & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

(a) a 2×2 image as shown in image (a) (10 pts)

(b) a 3×3 image as shown in image (b) (10 pts)

1	2
3	4

(a)

0	0	0
0	1	0
0	0	0

(b)

Setup: Convolution Formula

The convolution is defined as:

$$g(x, y) = \sum_{s=-1}^{1} \sum_{t=-1}^{1} w(s, t)\, f(x - s, y - t)$$

where $f(x, y) = 0$ outside the image (zero-padding). For a $M \times N$ image convolved with a 3×3 mask, the **full** output size is $(M + 2) \times (N + 2)$.
The mask w with indices $s, t \in \{-1, 0, 1\}$:

$$w(-1, -1) = -1,\ w(-1, 0) = 0,\ w(-1, 1) = -1$$

$$w(0, -1) = 0,\ w(0, 0) = 2,\ w(0, 1) = 0$$

$$w(1, -1) = 1,\ w(1, 0) = 0,\ w(1, 1) = 1$$

Part (a): 2×2 Image — Full 4×4 Output

Image: $f(0,0) = 1$, $f(0,1) = 2$, $f(1,0) = 3$, $f(1,1) = 4$. All other $f = 0$.
Output indices: $x \in \{-1, 0, 1, 2\}$, $y \in \{-1, 0, 1, 2\}$.

Row $x = -1$:

$$\begin{aligned}
g(-1,-1) &= w(-1,-1) \cdot f(0,0) = (-1)(1) = -1 \\
g(-1,\ 0\ ) &= w(-1,-1) \cdot f(0,1) = (-1)(2) = -2 \\
g(-1,\ 1\ ) &= w(-1,1) \cdot f(0,0) = (-1)(1) = -1 \\
g(-1,\ 2\ ) &= w(-1,1) \cdot f(0,1) = (-1)(2) = -2
\end{aligned}$$

Row $x = 0$:

$$\begin{aligned}
g(0,-1) &= w(-1,-1) \cdot f(1,0) = (-1)(3) = -3 \\
g(0,\ 0\ ) &= w(-1,-1) \cdot f(1,1) + w(0,0) \cdot f(0,0) = (-1)(4) + (2)(1) = -2 \\
g(0,\ 1\ ) &= w(-1,1) \cdot f(1,0) + w(0,0) \cdot f(0,1) = (-1)(3) + (2)(2) = 1 \\
g(0,\ 2\ ) &= w(-1,1) \cdot f(1,1) = (-1)(4) = -4
\end{aligned}$$

Row $x = 1$:

$$\begin{aligned}
g(1,-1) &= w(1,-1) \cdot f(0,0) = (1)(1) = 1 \\
g(1,\ 0\ ) &= w(1,-1) \cdot f(0,1) + w(0,0) \cdot f(1,0) = (1)(2) + (2)(3) = 8 \\
g(1,\ 1\ ) &= w(1,1) \cdot f(0,0) + w(0,0) \cdot f(1,1) = (1)(1) + (2)(4) = 9 \\
g(1,\ 2\ ) &= w(1,1) \cdot f(0,1) = (1)(2) = 2
\end{aligned}$$

Row $x = 2$:

$$\begin{aligned}
g(2,-1) &= w(1,-1) \cdot f(1,0) = (1)(3) = 3 \\
g(2,\ 0\ ) &= w(1,-1) \cdot f(1,1) = (1)(4) = 4 \\
g(2,\ 1\ ) &= w(1,1) \cdot f(1,0) = (1)(3) = 3 \\
g(2,\ 2\ ) &= w(1,1) \cdot f(1,1) = (1)(4) = 4
\end{aligned}$$

Part (a) Result Grid

-1	-2	-1	-2
-3	-2	1	-4
1	8	9	2
3	4	3	4

Verification: Sum of output $= -1-2-1-2-3-2+1-4+1+8+9+2+3+4+3+4 = 20$.
Sum of mask $= 2$, sum of image $= 10$. Product: $2 \times 10 = 20$. ✓

Part (b): 3×3 Image — Full 5×5 Output

Image: $f(1,1) = 1$, all other $f(i,j) = 0$. This is a unit impulse at position $(1,1)$.
Since the only nonzero pixel is $f(1,1) = 1$, each output value $g(x,y)$ has at most one nonzero term in the convolution sum, occurring when $x - s = 1$ and $y - t = 1$, i.e., $s = x - 1$, $t = y - 1$:

$$g(x,y) = w(x-1,\, y-1) \cdot f(1,1) = w(x-1,\, y-1)$$

This is nonzero only when $x - 1 \in \{-1, 0, 1\}$ and $y - 1 \in \{-1, 0, 1\}$, i.e., $x \in \{0,1,2\}$ and $y \in \{0,1,2\}$. All border positions in the 5×5 output are zero.
The central 3×3 block of the output is simply the filter w itself:

Part (b) Result Grid

0	0	0	0	0
0	-1	0	-1	0
0	0	2	0	0
0	1	0	1	0
0	0	0	0	0

Verification: Sum of output $= -1 - 1 + 2 + 1 + 1 = 2$. Sum of mask $= 2$, sum of image $= 1$. Product: $2 \times 1 = 2$. ✓
The convolution of a unit impulse with a filter produces the filter itself. This is the fundamental property of the impulse response.

Results

(a) The full 4×4 convolution output for the 2×2 image is:

$$g = \begin{bmatrix} -1 & -2 & -1 & -2 \\ -3 & -2 & 1 & -4 \\ 1 & 8 & 9 & 2 \\ 3 & 4 & 3 & 4 \end{bmatrix}$$

(b) The full 5×5 convolution output for the 3×3 impulse image is:

$$g = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The impulse response equals the filter w itself, embedded in a border of zeros. ■